

Data Management and Transformation

EVR 628- Intro to Environmental Data Science

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Table of contents

Exercise 1: Reading and transforming data	1
Part A) Obtaining data from the wild	2
Part B) Reading data	2
Part C) Renaming columns with <code>clean_names()</code> and <code>rename()</code>	3
Part D) Filtering rows with <code>filter()</code>	4
Part E) Creating new columns with <code>mutate()</code>	5
Part F) Calculating group summaries with <code>group_by()</code> and <code>summarize()</code>	5
Part G) Visualize the data and answer the question	6
Extra exercises for you to practice	7

Exercise 1: Reading and transforming data

You are the lead data analyst at an NGO that works to promote sustainable fisheries. Your boss asked you what sounds like a simple query:

How much money have tuna purse seiners made since 2000 when fishing for bigeye tuna (*Thunnus obesus*) in the Eastern Pacific Ocean?

Let's make some assumptions that will help us answer this question:

1. We will interpret “making money” as revenue, not profits (profits are revenue minus costs)
2. The market price of tuna since 2000 has remained relatively stable, at around US\$2/Kg (See Sibert et al. (2012))

Part A) Obtaining data from the wild

How to find the data:

1. Go to iattc.org
2. In the top menu, **hover** over DATA
3. Click on “Public domain”
4. You will be taken to a page titled “Public domain data for download”
5. We will use “EPO total estimated catch by year, flag, gear, species”
6. Click on `CatchByFlagGear.zip` to the right of the table to prompt a download
7. Save the zip file to **inside** your EVR628 project at: `data/raw/`¹
8. Using your finder / explorer, navigate to `EVR628/data/raw/` and unzip the `CatchByFlagGear.zip` file

! Windows users

You might have to click a button called “Extract” in the top of your explorer window.

9. You will get a new, normal folder called `CatchByFlagGear` under `EVR628/data/raw/`
10. Read the PDF file called `CarchByFlagGear`, which contains some documentation

Part B) Reading data

0. **Put your post-it up**
1. Start a new script called `tuna_analysis` and save it to your `scripts/03_analysis` folder
2. Add a comment header and outline, and load the `tidyverse` package at the top of your script
3. Use the `read_csv()` function to load the new data and assign it to an object called `tuna_data`
4. What are the existing column names?²
5. **Remove your post-it when you are done**

```
# Load packages
library(tidyverse)
library(janitor)

# Load the data
tuna_data <- read_csv("data/raw/CatchByFlagGear/CatchByFlagGear1918-2024.csv")
```

¹If your web browser didn't allow you to specify the download folder, your file is likely in the “Downloads” folder. Navigate to the downloads folder and copy the zipped folder to the `data/raw/` folder.

²Hint: use the `colnames()` function

```
# Check colnames after cleaning names
colnames(tuna_data)
```

```
[1] "AnoYear"          "BanderaFlag"      "ArteGear"         "EspeciesSpecies"
[5] "t"
```

Part C) Renaming columns with `clean_names()` and `rename()`

0. Put your post-it up

1. In your *console*, install the `janitor` package using `install.packages("janitor")`
2. Load `janitor` at the top of your script, run that line, and then read the documentation for the `clean_names()` function
3. Modify your code for the `tuna_data` object so that you pipe into `clean_names()` after reading the data
4. What are the new column names?
5. Extend the pipeline above so that we can use the `rename()` function. Rename the columns so that we only retain the English portion of the name. Let's also rename `t` as `catch`
6. Remove your post-it when you are done

```
# Load packages
library(tidyverse)
library(janitor)

# Load data
tuna_data <- read_csv("data/raw/CatchByFlagGear/CatchByFlagGear1918-2024.csv") |>
  # Clean column names
  clean_names() |>
  # Rename some columns
  rename(year = ano_year,
         flag = bandera_flag,
         gear = arte_gear,
         species = especies_species,
         catch = t)

# Check column names
colnames(tuna_data)
```

```
[1] "year"    "flag"    "gear"    "species" "catch"
```

Congratulations, you now have a tidy data set with which we can work! The next steps are to keep the data we care about, calculate revenues, and then calculate summaries. Let's do that.

Part D) Filtering rows with `filter()`

0. Put your post-it up

1. What are the unique species represented in the data?
2. What are the fishing gears represented in the data?
3. Does the documentation say what these codes are?³
4. What is the species code for bigeye tuna?
5. What is the gear code for purse seine?
6. Create a new object called `ps_tuna_data` that takes the `tuna_data` and filters it to retain data for:
 - a. bigeye tuna
 - b. caught by tuna purse seiners
 - c. since 2000

7. Remove your post-it when you are done

```
# Check unique values for the species column (using a pipe)
tuna_data$species |> unique()
```

```
[1] "SKJ" "YFT" "SWO" "BZX" "ALB" "PBF" "BET" "BLM" "BUM" "MLS" "CGX" "MZZ"
[13] "BKJ" "BIL" "SKH" "TUN" "DOX" "SFA" "SSP" "SRX" "BXQ"
```

```
# Check unique values for the gear column, without a pipe
unique(tuna_data$gear)
```

```
[1] "LP" "PS" "UNK" "HAR" "LTL" "RG" "LL" "GN" "OTR" "LHP" "TX"
```

```
# Create a new data set called ps_tuna_data after filtering
ps_tuna_data <- tuna_data |>
  filter(species == "BET",
         gear == "PS",
         year >= 2000)
```

³Hint: There is a cryptic link to the [reference codes](#)

Part E) Creating new columns with mutate()

0. Put your post-it up

1. Modify the `ps_tuna_data` pipeline to create a new column called `revenue` that calculates the revenue generated by selling the catch⁴
2. Make sure you calculate revenues in Millions of USD
3. **Remove your post-it when you are done**

```
# Create a new data set called ps_tuna_data after filtering
ps_tuna_data <- tuna_data |>
  filter(species == "BET", # Retain BET values only
         gear == "PS",    # Retain PS values only
         year >= 2000) |> # Retain data from 2000 onwards
  mutate(revenue = catch * 1000 * 2 / 1e6) # Calculate revenue
```

Part F) Calculating group summaries with group_by() and summarize()

0. Put your post-it up

1. The data right now report catch at the year-by-flag level. Modify the `ps_tuna_data` pipeline so that we have total catch and revenue by year.⁵
2. **Remove your post-it when you are done**

```
ps_tuna_data <- tuna_data |>
  filter(species == "BET", # Retain BET values only
         gear == "PS",    # Retain PS values only
         year >= 2000) |> # Retain data from 2000 onwards
  mutate(revenue = catch * 1000 * 2 / 1e6) |> # Calculate revenue
  group_by(year) |> # Specify that I am grouping by year
  # Tell summarize that I want to collapse the catch column by summing all its values
  summarize(catch = sum(catch),
            revenue = sum(revenue)) # Same, but for revenues
```

! Important

During class we only calculated total revenue. The above code calculates total revenue AND total catch.

⁴Hint: If I catch 10 kilos and the price per kilo is US\$2, then I make US\$20 because $10 * 2 = 20$

⁵Hint: You will need to use the `group_by`, `summarize()`, and `sum()` functions.

Part G) Visualize the data and answer the question

Remember, the question was:

How much money have tuna purse seiners made since 2000 when fishing for bigeye tuna (*Thunnus obesus*)?

The question is ambiguous because one could answer “They have made X M USD since 2000” or “Every year since 2000, they have made Y M USD per year.” So let’s get both:

0. Put your post-it up

1. What is the total revenue?⁶
2. What is the average annual revenue?⁷
3. Build a time-series showing revenues by year. Make sure to correctly label the axis and include a title and caption describing the figure and the data source, respectively.
4. Remove your post-it when you are done.

```
# Get total revenue
sum(ps_tuna_data$revenue)
```

```
[1] 3139.372
```

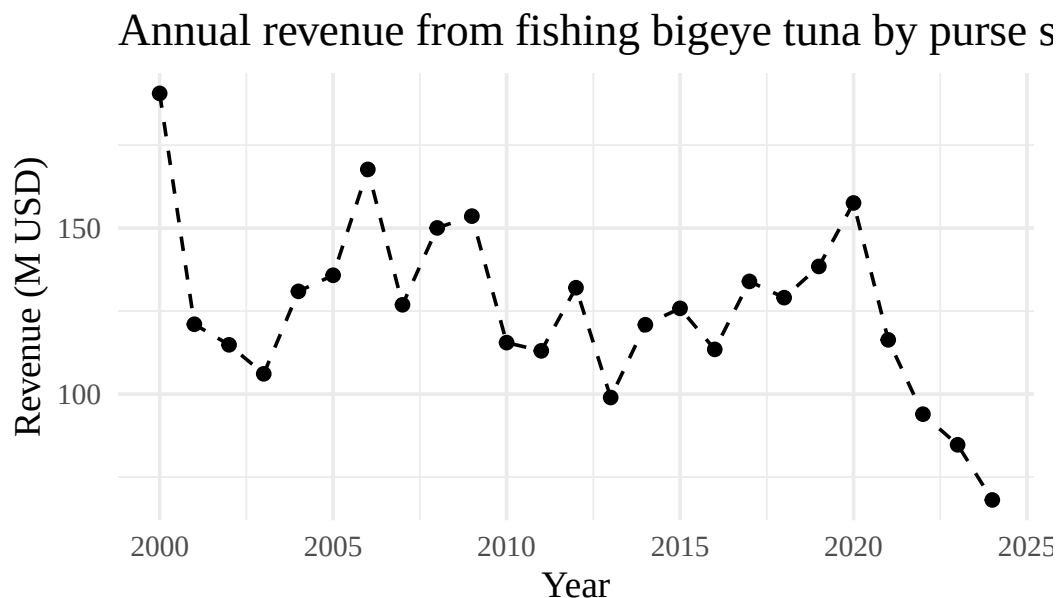
```
# Get mean annual revenue
mean(ps_tuna_data$revenue)
```

```
[1] 125.5749
```

```
# Build plot
ggplot(data = ps_tuna_data,                                # Specify my data
       mapping = aes(x = year, y = revenue)) +             # And my aesthetics
geom_line(linetype = "dashed") +                           # Add a dashed line
geom_point() +                                              # With points on top
labs(x = "Year",                                           # Add some labels
     y = "Revenue (M USD)",
     title = "Annual revenue from fishing bigeye tuna by purse seine vessels",
     caption = "Data come from the IATTC") +
# Modify the theme
theme_minimal(base_size = 14,                             # Font size 14
              base_family = "Times")                      # Font family Times
```

⁶Hint: Use \$ and sum()

⁷Hint: Use \$ and mean()

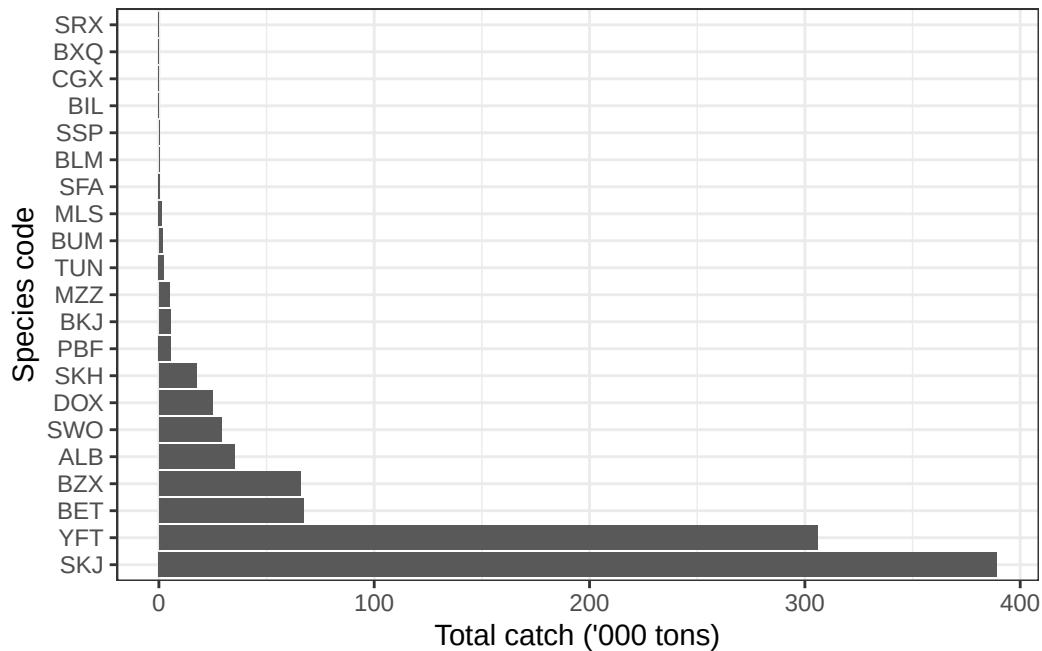


Extra exercises for you to practice

1. Make a figure showing total catch by species during 2023

```
# Build a new data.frame that has catch by species (in thousand tons)
catch_2023 <- tuna_data |>
  filter(year == 2023) |>
  group_by(species) |>
  summarize(total_catch = sum(catch) / 1e3)

# Now build the figure
ggplot(data = catch_2023,
       aes(x = fct_reorder(species, total_catch, .desc = T), # I am using this fct_reorder to
           y = total_catch)) +
  geom_col() +
  labs(x = "Species code",
       y = "Total catch ('000 tons)") +
  coord_flip() +
  theme_bw()
```



2. Which country caught the most tuna during 2020?

```
# There is more than one way to do this one
# Option 1

# Build a data that has total catch by country in 2020
country_catch <- tuna_data |>
  filter(year == 2020) |> # Retain only data from 2020
  group_by(flag) |> # Get total catch by flag (i.e. sum catch across all species)
  summarize(total_catch = sum(catch))

# And then we can use brackets, dollar signs, and boolean operators to extract the flag
country_catch$flag[country_catch$total_catch == max(country_catch$total_catch)]
```

[1] "ECU"

```
# Option2: The way I haven't shown you
tuna_data |>
  filter(year == 2020) |>
  group_by(flag) |>
  summarize(total_catch = sum(catch)) |> # Up until here, the pipeline is the same
  arrange(desc(total_catch)) |> # Then I use the arrange function to sort the data in descending order
```

```
head(1) |> # I then retain only the first row, which now _should_ contain the data I want
pull(flag) # This is a "tidy" version of using a dollar sign to extract a column
```

```
[1] "ECU"
```

3. For each species, identify the year in which catch was at it's maximum

```
# Option 1: With what you already know
tuna_data |>
  group_by(year, species) |>
  summarize(total_catch = sum(catch)) |> # We firrst calculate total catch by species and year
  group_by(species) |> # Then we group by species
  filter(total_catch == max(total_catch)) |> # And use the filter function. Since the data are sorted by year
  select(species, year_of_max_catch = year) |> # And we keep the columns we care about
  arrange(species)
```

```
# A tibble: 21 x 2
# Groups:   species [21]
  species year_of_max_catch
  <chr>         <dbl>
1 ALB             2014
2 BET             2000
3 BIL             2013
4 BKJ             2016
5 BLM             1973
6 BUM             1963
7 BXQ             2019
8 BZX             2023
9 CGX             1983
10 DOX            2009
# i 11 more rows
```

```
# Option 2: Using slice_max
tuna_data |>
  group_by(year, species) |>
  summarize(total_catch = sum(catch)) |>
  group_by(species) |>
  slice_max(total_catch) |>
  select(species, year_of_max_catch = year) |>
  arrange(species)
```

```
# A tibble: 21 x 2
# Groups:   species [21]
  species year_of_max_catch
  <chr>      <dbl>
1 ALB          2014
2 BET          2000
3 BIL          2013
4 BKJ          2016
5 BLM          1973
6 BUM          1963
7 BXQ          2019
8 BZX          2023
9 CGX          1983
10 DOX         2009
# i 11 more rows
```

4. How many species have been caught by Mexican-flagged vessels since 2000?

```
# Option 1: A pipeline that ends in a vector, with unique and length
# Start from tuna_data
tuna_data |>
  filter(flag == "MEX") |> # Retain observations associated with Mexico
  pull(species) |> # Pull the species column away from the data.frame, at this point we have
  unique() |> # Get a unique list of species
  length() # Count the number of unique species
```

```
[1] 21
```

```
# Alternatively, retain the data.frame structure
tuna_data |>
  filter(flag == "MEX") |> # Retain observations associated with Mexico
  group_by(flag) |>
  summarize(n_species = n_distinct(species)) # The n_distinct() function is a tidy version of
```

```
# A tibble: 1 x 2
  flag n_species
  <chr>   <int>
1 MEX       21
```

Sibert, John, Inna Senina, Patrick Lehodey, and John Hampton. 2012. "Shifting from Marine Reserves to Maritime Zoning for Conservation of Pacific Bigeye Tuna (*Thunnus Obesus*)."
Proc. Natl. Acad. Sci. U. S. A. 109 (October): 18221–25.